

Report 9 - Core Component Model Families Overview

Literature, Mathematics, OpenHPL Interpretation, and Julia Implementation

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Abstract

OpenHPLjl v0.1.0-alpha.1 established a core component baseline. This report surveys the model families that should exist inside each current component folder before the next implementation milestone. For each family it summarizes the engineering concept, mathematical hierarchy, OpenHPL/OpenIPSL analogue, Julia/ModelingToolkit implementation route, and recommended next fidelity level.

1 Updated Engineering Context

The current chain is

Reservoir → RigidPipe → SurgeTank → Turbine → Shaft → Generator → Grid.

OpenHPL 3.0.1 similarly separates waterways, electro-mechanical models, interfaces and functions. Its User Guide states that waterway components are based on mass and momentum balances, includes elastic/compressible waterway models, mechanistic and lookup-table turbines, and supports coupling to OpenIPSL for electrical-system models [1].

The next milestone is therefore a model-family expansion rather than simply adding more isolated files.

2 Generalized Model-Hierarchy Concept

For every folder, the preferred fidelity ladder is

boundary/algebraic → lumped dynamic → nonlinear physics-based → distributed/high-order.

The connector interface should remain stable while the internal equations change.

3 Literature Model Families Overview

Folder	Concept	Model families	Mathematical family
Interfaces	Domain coupling and conservation	Hydraulic effort/flow; rotational speed/torque; reduced electrical frequency/power; later voltage/current phasor ports	Across variables are equated; flow variables sum to zero. ModelingToolkit uses <code>connect = Flow</code> for conserved connector variables [10].
Reservoirs	Hydraulic storage and boundary head	Infinite reservoir; constant-area reservoir; nonlinear area-volume reservoir; reservoir-channel/open-channel; hydrology-driven inflow	$\dot{V} = Q_{in} - Q_{out}$ and $A(H)\dot{H} = Q_{net}$. OpenHPL includes Reservoir and ReservoirChannel models [2].
FrictionModels	Hydraulic dissipation	No loss; identified $RQ Q $; constant Darcy factor; laminar; Haaland; Swamee-Jain; Colebrook; transition; unsteady friction	$H_f = f_D(L/D)Q Q /(2gA^2)$ and $Re = 4 Q /(\pi D\nu)$. OpenHPL distinguishes laminar, transition and turbulent Darcy-factor regimes [3].
RigidPipes	Water inertia and transient head	Quasi-steady conduit; rigid water-column ODE; elastic water-hammer PDE; method of characteristics; finite-volume/KP model	Rigid: $\frac{L}{gA}\dot{Q} = H_u - H_d - H_f$. Elastic: coupled continuity and momentum PDEs in $H(x, t), Q(x, t)$ [8].

Folder	Concept	Model families	Mathematical family
SurgeTanks	Local storage and hydraulic oscillation	Simple; throttled; sharp-orifice; air-cushion; variable-area; differential/combined	$A_s(H)\dot{H}_s = Q_{net}$ plus throttle/orifice loss. Air-cushion variants add a gas pressure-volume relation. OpenHPL provides four types [4].
Turbines	Hydraulic-to-mechanical conversion	Ideal; simple gate/valve; constant-efficiency; efficiency lookup; Hill-chart lookup; mechanistic Francis; Pelton; Kaplan; servo-coupled	$P_m = \rho g \eta Q H$, reduced $Q = K_q y \sqrt{H}$, then $Q = f(H, \omega, y)$ and $\eta = f(H, Q, \omega, y)$. OpenHPL has simple, Francis and Pelton models [5, 7].
Shafts	Mechanical inertia and torsion	Rigid coupling; single inertia; two-mass torsional; multi-mass flexible shaft; friction/backlash variants	$J\dot{\omega} = \sum \tau - D\Delta\omega$. Two-mass models add torsional stiffness and damping. Modelica Rotational provides inertia, spring and spring-damper components [11].
Generators	Electromechanical conversion	Ideal converter; classical/swing; third/fourth-order transient-EMF; sixth-order dq; salient/round rotor; saturation and excitation attachments	$P_m = \tau\omega$; swing $M\dot{\omega} = P_m - P_e - D\Delta\omega$, $\dot{\delta} = \omega - \omega_s$. Higher orders add transient/subtransient EMF states. OpenIPSL includes GENCLS, GENROU, GENSAL and related families [12].
Grids	Electrical boundary and network dynamics	Infinite bus; aggregate finite-inertia grid; SMIB network; multi-machine phasor network; later EMT if required	Infinite bus: $\omega = \omega_0$. Aggregate grid: swing/load-damping equation. SMIB: power-angle relation. Multi-machine: nodal DAEs. OpenIPSL GENCLS can represent a classical machine or infinite bus and has SMIB validation systems [13, 14].

4 Mathematical Families

The current folders reduce to five mathematical families.

4.1 Storage

Reservoirs and surge tanks use

$$\dot{V} = \sum Q, \quad A(H)\dot{H} = \sum Q.$$

4.2 Momentum

Rigid waterways use

$$\frac{L}{gA}\dot{Q} = H_u - H_d - H_f(Q).$$

Elastic waterways introduce

$$\frac{\partial H}{\partial t} + \frac{a^2}{gA} \frac{\partial Q}{\partial x} = 0,$$

$$\frac{\partial H}{\partial x} + \frac{1}{gA} \frac{\partial Q}{\partial t} + f \frac{Q|Q|}{2gDA^2} = 0.$$

Recent hydropower reviews distinguish rigid and elastic water-column models and describe MOC, FDM, FVM and FEM solution approaches [8].

4.3 Constitutive and lookup models

Friction and turbine maps are mainly algebraic:

$$H_f = \phi(Q, p), \quad Q = f_Q(H, \omega, y), \quad \eta = f_\eta(H, Q, \omega, y).$$

Hill charts are a natural representation of turbine efficiency and operating regions [9].

4.4 Rotational dynamics

For one inertia,

$$J\dot{\omega} = \tau_m - \tau_e - D\Delta\omega.$$

For two inertias,

$$\begin{aligned} J_1\dot{\omega}_1 &= \tau_t - K_t(\theta_1 - \theta_2) - D_t(\omega_1 - \omega_2), \\ J_2\dot{\omega}_2 &= K_t(\theta_1 - \theta_2) + D_t(\omega_1 - \omega_2) - \tau_e. \end{aligned}$$

4.5 Electromechanical/network DAEs

A classical generator uses

$$\begin{aligned} \dot{\delta} &= \omega - \omega_s, \\ M\dot{\omega} &= P_m - P_e - D(\omega - \omega_s), \end{aligned}$$

while the network supplies algebraic electrical-power equations. OpenIPSL documents GENCLS as a classical generator that can also represent an infinite bus depending on inertia parameterization [13].

5 OpenHPL and Modelica Interpretation

OpenHPL already provides a useful fidelity map:

- Waterways: reservoir, simple pipe, elastic penstock, KP-scheme penstock, open channel, fittings, gates and surge tanks [2].
- Surge tanks: simple, air-cushion, throttle-valve and sharp-orifice [4].
- Turbines: simple turbine with constant or lookup efficiency plus mechanistic Francis and Pelton models [6, 7].
- Electrical integration: OpenHPL is designed to interoperate with OpenIPSL [1].
- Shafts: Modelica Rotational supplies inertias, springs, dampers and spring-dampers suitable for rigid and flexible drive-train models [11].

6 Julia / ModelingToolkit Implementation Pattern

Every model family should follow the same development pattern:

```
literature model
-> governing equations
-> parameter table
-> stable connector interface
-> ModelingToolkit component
-> analytical unit test
-> reference experiment
-> higher-fidelity sibling model
```

Recommended source expansion:

```
Reservoirs/
  NonlinearReservoir.jl
  ReservoirChannel.jl

Waterways/RigidPipes/
  ElasticPenstock.jl
  MOCWaterway.jl

Waterways/SurgeTanks/
  ThrottledSurgeTank.jl
  AirCushionSurgeTank.jl
  OrificeSurgeTank.jl

Turbines/
  TurbineLookup.jl
  FrancisTurbine.jl
  PeltonTurbine.jl
```

```

KaplanTurbine.jl

Mechanical/Shfts/
TwoMassShaft.jl

Electrical/Generators/
ClassicalGenerator.jl
FourthOrderGenerator.jl
SixthOrderGenerator.jl

Electrical/Grids/
AggregateGrid.jl
SMIBGrid.jl
MultiMachineGrid.jl

```

7 Implementation Priority Table

Family	Baseline	Implement next	Reason
Interfaces	Three current ports	Add voltage/current phasor port only when needed	Avoid premature electrical detail.
Reservoirs	Infinite + constant-area	Nonlinear $A(H)$ reservoir	Useful and low complexity.
Friction	Current registry	Transition blend + calibration support	Important for measurement validation.
RigidPipes	Rigid column	Elastic penstock / MOC or FVM prototype	Required for water hammer.
SurgeTanks	Simple	Throttled + air-cushion	Important for oscillation damping and design.
Turbines	Ideal + gate	Lookup/Hill chart, then Francis	Highest value for real-plant validation and FCR.
Shafts	Lumped inertia	Two-mass shaft	Natural torsional extension.
Generators	Ideal converter	Classical generator, then fourth-order	Required for rotor-angle/frequency dynamics.
Grids	Infinite grid	SMIB grid + aggregate finite-inertia grid	Required for frequency/stability studies.

8 Validation Strategy

Each new model should pass:

1. equation-level analytical or conservation test;
2. component step/ramp experiment;
3. limiting-case comparison with a simpler sibling model;
4. connected-system structural and initialization test.

Examples include:

- elastic penstock approaching rigid-pipe behaviour for slow transients;
- throttled surge tank approaching simple surge tank as throttle loss tends to zero;
- lookup turbine approaching constant-efficiency behaviour near one operating point;
- two-mass shaft approaching lumped shaft for large torsional stiffness;
- classical generator approaching stiff-grid behaviour as inertia becomes very large, consistent with GENCLS [13].

9 Expected Outputs

The expanded library should expose:

- Hydraulic: H, Q, H_f , pressure, surge level and wave states.
- Turbine: Q, H, y, η, P_m, T_m and map coordinates.
- Mechanical: ω, θ , shaft torque and torsional deflection.
- Generator: δ, ω, P_m, P_e , transient EMFs and currents where applicable.
- Grid: frequency, angle, active/reactive power and bus voltage where applicable.

10 Engineering Interpretation and Next Milestone

The roadmap becomes:

Layer 1: Core components - completed in v0.1.0-alpha.1

Layer 2: Model-family expansion

Layer 3: Plant validation, governor, droop, AGC and FCR

Recommended next implementation sequence:

TurbineLookup $\rightarrow$ ClassicalGenerator $\rightarrow$ SMIBGrid $\rightarrow$ TwoMassShaft $\rightarrow$ ThrottledSurgeTank $\rightarrow$ ElasticPenstock

This sequence produces a useful connected plant early while preparing the hydraulic side for higher-frequency transient studies.

Conclusion

The package now has a literature-guided model-family roadmap. The main architectural rule is to preserve connectors while increasing internal model fidelity. This lets OpenHPLjl support both reduced control studies and higher-fidelity hydropower transient studies without duplicating system architecture.

References

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